بِسۡمِ ٱللَّهِ ٱلرَّحۡمَٰنِ ٱلرَّحِيمِ

کیا آپ نے آج درود شریف پڑھا

نہیں تو ابھی پڑھ لیجئے

صلی اللہ علیہ و سلم

لوح قرآنی

الٓمّٓ	حٰمٓ عٓسٓقٓ	نٓ
الٓمٓصٓ	حٰمٓ	یٰسٓ
کٓھٰیٰعٓصٓ	قٓ	اٰمین

اس کے دیکھنے والوں کی سب مشکلیں آسان ہوجائیں گی صبح اس کو دیکھ کر جو کام شروع کیا جائیگا' پورا ہوگا' اور غیبی طریقوں سے رزق آنے لگے گا!

Question # 6 of 10 (Start time: 08:45:01 AM, 04 September 2022)

If X and Y are independent random variables and a and b are constants, then Var(aX+bY)is equal to

Select the correct option

- ( ) a^2 Var(X)+ b^2 Var(Y)
- (•) aVar(X)+ bVar(Y)
- ( ) Var(aX)+ Var(bY)
- ( ) (a+b)[Var(X)+ Var(Y)]

Email: waqasahmeduog@gmail.com

Question # 9 of 10 (Start time: 08:52:35 AM, 04 September 2022)

Conditional probability of B given A is defined as ....

Select the correct option

- ○ None of these
- ● $\frac{P(A \cap B)}{P(A)}$
- ○ $\frac{P(A \cap B)}{P(B)}$
- ○ $P(A \cap B)$

Email: waqasahmeduog@gmail.com

Question # 1 of 10 (Start time: 08:26:44 AM, 04 September 2022)

An event is a subset of?

Select the correct option

- ( ) Experiment
- (•) sample space

Email: waqasahmeduog@gmail.com

MTH202 - Discrete Mathematics (Quiz No.4)

Question # 2 of 10 (Start time: 08:27:00 AM, 04 September 2022)

If a pair of dice is tossed how many outcomes the sample space of the experiment will have?

Select the correct option

○	12
○	18
○	6
●	36

Email: waqasahmeduog@gmail.com

Question # 3 of 10 (Start time: 08:27:16 AM, 04 September 2022)

If a die is tossed how many outcomes the sample space of the experiment will have?

Select the correct option

- ( ) 5
- (•) 6
- ( ) 12
- ( ) 36

Email: waqasahmeduog@gmail.com

Question # 4 of 10 (Start time: 08:27:37 AM, 04 September 2022)

There are 5 girls students and 20 boys students in a class. How many students are there in total ?

Select the correct option

○	100
○	15
●	25
○	4

Email: waqasahmeduog@gmail.com

Question # 5 of 10 (Start time: 08:27:46 AM, 04 September 2022)

If A and B are two independent events then $P(A \cap B) = \ldots\ldots\ldots\ldots\ldots\ldots$

Select the correct option

- P(B)
- $P(A)P(B)$
- P(A)
- 1

Email: waqasahmeduog@gmail.com

Question # 7 of 10 (Start time: 08:29:12 AM, 04 September 2022)

Suppose that A and B are events in a sample space S. If A and B are disjoint, could $P(A)=0.6$ and $P(B)=0.5$?

Select the correct option

- [x] No
- [ ] Yes

Email: waqasahmeduog@gmail.com

MTH202 - Discrete Mathematics (Quiz No.4)

Question # 8 of 10 (Start time: 08:29:38 AM, 04 September 2022)

If two light bulbs are chosen at random from 5 bulbs of which 3 are defective, then which of the following is the probability t

Select the correct option

- 2/10
- 1/10
- 2/8
- 2/7

Email: waqasahmeduog@gmail.com

Question # 9 of 10 (Start time: 08:29:49 AM, 04 September 2022)

The expected value E(X) is obtained by multiplying each value of x with its probability and taking the sum.

Select the correct option

- ( ) False
- (•) True

Email: waqasahmeduog@gmail.com

Question # 10 of 10 (Start time: 08:30:10 AM, 04 September 2022)

If A and B are finite sets then the mathematical representation of inclusion-exclusion principle is ....

Select the correct option

- ( ) $n(A \cup B) = n(A) + n(B)$
- (•) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- ( ) $n(A \cup B) = n(A) + n(B) - n(A \cup B)$
- ( ) Non of these

Email: waqasahmeduog@gmail.com

Question # 3 of 10 (Start time: 08:33:54 AM, 04 September 2022)

If A and B are disjoint finite sets then

$$n(A \cup B) = \_\_\_\_\_\_.$$

Select the correct option

- ( ) $n(A) + n(B) - n(A \cap B)$
- ( ) $n(A) - n(B)$
- ( ) $n(A) + n(B) + n(A \cap B)$
- (•) $n(A) + n(B)$

Email: waqasahmeduog@gmail.com

There are 5 girls students and 20 boys students in a class. How many students are there in total ?

Select the correct option

- ( ) 4
- (•) 25
- ( ) 100
- ( ) 15

Email: waqasahmeduog@gmail.com

Question # 5 of 10 (Start time: 08:34:25 AM, 04 September 2022)

If X and Y are random variables, then E(X-Y)is equal to

Select the correct option

- ○ E(X)+E(Y)
- ○ E(X+Y)
- ○ E(X-Y)
- ● E(X)-E(Y)

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Question # 6 of 10 (Start time: 08:34:52 AM, 04 September 2022)

If A and B are finite sets then the mathematical representation of inclusion-exclusion principle is ....

Select the correct option

- ( ) $n(A \cup B) = n(A) + n(B)$
- ( ) Non of these
- (•) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- ( ) $n(A \cup B) = n(A) + n(B) - n(A \cup B)$

Email: waqasahmeduog@gmail.com

Question # 8 of 10 (Start time: 08:35:13 AM, 04 September 2022)

Let $E(X^2) = 6\,,\ \mu^2 = 2,$ then standard deviation is ....

Select the correct option

- ( ) 8
- ( ) 4
- ( ) -2
- (•) 2

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Question # 9 of 10 (Start time: 08:35:26 AM, 04 September 2022)

If A and B are finite sets then

$$n(A \cup B) = \_\_\_\_\_\_.$$

Select the correct option

- ( ) $n(A) + n(B)$
- ( ) $n(A) + n(B) + n(A \cap B)$
- ( ) $n(A) + n(B) - n(A \cup B)$
- (•) $n(A) + n(B) - n(A \cap B)$

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MTH202 - Discrete Mathematics (Quiz No.4)

Question # 10 of 10 (Start time: 08:36:24 AM, 04 September 2022)

If a pair of dice is tossed how many outcomes the sample space of the experiment will have?

Select the correct option

- ○ 18
- ○ 6
- ◉ 36
- ○ 12

Email: waqasahmeduog@gmail.com

Question # 1 of 10 (Start time: 08:40:49 AM, 04 September 2022)

A Random variable is also called a

Select the correct option

- (selected) Chance Variable
- Constant

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Question # 3 of 10 (Start time: 08:42:40 AM, 04 September 2022)

If X and Y are independent random variables, then E(XY) is equal to ___________.

Select the correct option

- ○ YE(X)
- ○ E(XY)
- ● E(X)E(Y)
- ○ XE(Y)

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MTH202 - Discrete Mathematics (Quiz No.4)

Question # 1 of 10 (Start time: 06:12:36 PM, 04 September 2022)

Who gave the first definition of the probability as the number of successful outcomes divided by the number of total outco

Select the correct option

none

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Leonhard Euler

Blaise Pascal

Laplace

Question # 1 of 10 (Start time: 06:12:36 PM, 04 September 2022)

Who gave the first definition of the probability as the number of successful outcomes divided by the number of total outco

Select the correct option

- none
- Leonhard Euler
- Blaise Pascal
- Laplace

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MTH202 - Discrete Mathematics (Quiz No.4)

Question # 2 of 10 (Start time: 06:14:20 PM, 04 September 2022)

Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog ?

Select the correct option

- 24
- 6
- 16
- 46

Email: waqasahmeduog@gmail.com

Question # 3 of 10 (Start time: 06:15:44 PM, 04 September 2022)

If A and B are two independent events then $P(A \cap B) = \ldots\ldots\ldots\ldots\ldots\ldots$

Select the correct option

- 1
- P(A)
- $P(A)P(B)$ (selected)
- P(B)

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Question # 4 of 10 (Start time: 06:16:02 PM, 04 September 2022)

How many bit strings of length two do not have two consecutive 1's ?

Select the correct option

- 4
- 3
- 2
- 1

Email: waqasahmeduog@gmail.com

Question # 5 of 10 (Start time: 06:16:18 PM, 04 September 2022)

Suppose that A and B are events in a sample space S. If A and B are disjoint, could $P(A)=0.6$ and $P(B)=0.5$?

Select the correct option

- ( ) Yes
- (•) No

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MTH202 - Discrete Mathematics (Quiz No.4)

Question # 6 of 10 (Start time: 06:16:39 PM, 04 September 2022)

An urn contains six red and nine blue balls. What is the probability that a ball chosen from the urn is blue?

Select the correct option

- ( ) 6/15
- (•) 9/15
- ( ) 15/9
- ( ) 15/6

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Question # 7 of 10 (Start time: 06:16:49 PM, 04 September 2022)

If A and B are disjoint sets then the mathematical representation of inclusion-exclusion principle is ....

Select the correct option

- $n(A \cup B) = n(A) + n(B) - n(A \cup B)$
- $n(A \cup B) = n(A) + n(B)$ (selected)
- None of these
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Email: waqasahmeduog@gmail.com

Question # 8 of 10 (Start time: 06:17:06 PM, 04 September 2022)

If B is a subset of A then

$$P(A|B) = 1$$

.

Select the correct option

- ( ) Flase
- (•) True

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Question # 9 of 10 (Start time: 06:17:22 PM, 04 September 2022)

What is the probability of the number of one head when two fair coins are tossed?

Select the correct option

- (selected) 2/4
- 1/4

لوح قرآنی

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الٓمٓصٓ	حٰمٓ	یٰسٓ
کٓھٰیٰعٓصٓ	قٓ	اٰمین

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MTH202 - Discrete Mathematics (Quiz No.3.)

Question # 1 of 10 (Start time: 04:16:23 PM, 19 August 2022)

GCD of (9, 27) will be ______.

Select the correct option

- ○ 3
- ○ 27
- ○ 5
- ● 9

Email: waqasahmeduog@gmail.com

Question # 10 of 10 (Start time: 06:17:34 PM, 04 September 2022)

Conditional probability of A given B is defined as ....

Select the correct option

- (selected) $\frac{P(A \cap B)}{P(B)}$
- $\frac{P(A \cap B)}{P(a)}$
- $P(A \cap B)$
- None of these

Email: waqasahmeduog@gmail.com

Question # 4 of 10 (Start time: 08:43:24 AM, 04 September 2022)

If the standard deviation of the data set is 4 then the value of the variance is ....

Select the correct option

- ( ) 4
- (•) 16
- ( ) 2
- ( ) 8

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Question # 9 of 10 (Start time: 06:10:13 PM, 03 September 2022)

If A and B are disjoint sets then the mathematical representation of inclusion-exclusion principle is ....

Select the correct option

- ( ) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- ( ) None of these
- (•) $n(A \cup B) = n(A) + n(B)$
- ( ) $n(A \cup B) = n(A) + n(B) - n(A \cup B)$

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Question # 10 of 10 (Start time: 06:10:56 PM, 03 September 2022)

Let A and B be subsets of U with n(A) = 12, n(B) = 15, n(A')=17, and n(A intersection B) = 8, then n(U)=______

Select the correct option

- [x] 29
- [ ] 27
- [ ] 20
- [ ] 35

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Question # 1 of 10 (Start time: 06:12:47 PM, 03 September 2022)

Conditional probability of A given B is defined as ....

Select the correct option

- (selected) $\frac{P(A\cap B)}{P(B)}$
- $\frac{P(A\cap B)}{P(a)}$
- None of these
- $P(A \cap B)$

Email: waqasahmeduog@gmail.com

Question # 2 of 10 (Start time: 06:13:54 PM, 03 September 2022)

A random variable is also called a Stochastic variable

Select the correct option

- [x] True
- [ ] False

Email: waqasahmeduog@gmail.com

Question # 3 of 10 (Start time: 06:14:49 PM, 03 September 2022)

What is the probability of getting a number greater than 5 when a dice is tossed?

Select the correct option

- ( ) 2/6
- ( ) 1/3
- ( ) 1/5
- (•) 1/6

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Question # 4 of 10 (Start time: 06:15:35 PM, 03 September 2022)

In which century the theory of probability was first developed?

Select the correct option

- ( ) 16th century
- ( ) 19th century
- ( ) 18th century
- (•) 17th century

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Question # 5 of 10 (Start time: 06:16:06 PM, 03 September 2022)

If A is any set and U is the universalset then

$$n(A') = \_\_\_\_\_\_.$$

Select the correct option

- $n(U) + n(U \cap A)$
- $n(U) - n(U \cup A)$
- $n(U) + n(U \cup A)$
- $n(U) - n(U \cap A)$

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Question # 6 of 10 (Start time: 06:17:34 PM, 03 September 2022)

An expected value of a random variable is equal to its mean.

Select the correct option

- (•) True
- ( ) False

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Question # 7 of 10 (Start time: 06:18:40 PM, 03 September 2022)

If X and Y are random variables, then E(aX)is equal to

Select the correct option

- E(aX)
- aX
- aE(X)
- None of these

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Question # 8 of 10 (Start time: 06:19:12 PM, 03 September 2022)

If A and B be events with P(A)=1/3, P(B)=1/4 and P(A intersection B)=1/6, then P(A U B)= _________ .

Select the correct option

- 2/3
- 1/2
- 5/12
- 1/24

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Question # 9 of 10 (Start time: 06:20:08 PM, 03 September 2022)

There are 5 girls students and 20 boys students in a class. How many students are there in total ?

Select the correct option

- 4
- 15
- 100
- 25

MTH202 - Discrete Mathematics (Quiz No.4)

Question # 1 of 10 (Start time: 06:26:48 PM, 03 September 2022)

If E(5)=5, then find arithmetic mean willl be

Select the correct option

- ( ) 10
- ( ) 1
- (•) 5
- ( ) 0

Email: waqasahmeduog@gmail.com

Question # 2 of 10 (Start time: 06:28:00 PM, 03 September 2022)

An event is a subset of?

Select the correct option

- (selected) sample space
- Experiment

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Question # 3 of 10 (Start time: 06:28:13 PM, 03 September 2022)

An urn contains six red and nine blue balls. What is the probability that a ball chosen from the urn is blue?

Select the correct option

- 9/15
- 6/15
- 15/9
- 15/6

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Question # 4 of 10 (Start time: 06:28:55 PM, 03 September 2022)

If A and B are disjoint sets then the mathematical representation of inclusion-exclusion principle i

Select the correct option

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B) = n(A) + n(B)$
- None of these
- $n(A \cup B) = n(A) + n(B) - n(A \cup B)$

Email: waqasahmeduog@gmail.com

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Question # 6 of 10 (Start time: 06:31:04 PM, 03 September 2022)

The event A ∪ B may be written as?

Select the correct option

- $A \cup B = (A \setminus B) \cap B$
- $A \cup B = (A \setminus B) - B$

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- $A \cup B = (A \setminus B)$
- $A \cup B = (A \setminus B) \cup B$

Question # 7 of 10 (Start time: 06:32:03 PM, 03 September 2022)

What is the probability of getting a number greater than 5 when a dice is tossed?

Select the correct option

- 2/6
- 1/5
- 1/3
- 1/6

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The sum of probabilities of all outcome is always equal to "0".

Select the correct option

- True
- False

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Question # 10 of 10 (Start time: 06:34:30 PM, 03 September 2022)

A boy have 3 red and 4 blue balls, What is the probability that a ball chosen from the boy is blue?

Select the correct option

- ( ) 4/7
- ( ) none
- (•) 3/7
- ( ) 1/7

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Question # 5 of 10 (Start time: 06:39:51 PM, 03 September 2022)

If A and B are disjoint finite sets then

$$n(A \cup B) = \_\_\_\_\_.$$

Select the correct option

- ○ $n(A) - n(B)$
- ● $n(A) + n(B)$
- ○ $n(A) + n(B) + n(A \cap B)$
- ○ $n(A) + n(B) - n(A \cap B)$

Email: waqasahmeduog@gmail.com

Question # 9 of 10 (Start time: 08:46:16 AM, 04 September 2022)

If A and B are two independent events then $P(A \cap B) = \ldots\ldots\ldots\ldots\ldots\ldots$

Select the correct option

- (selected) $P(A)P(B)$
- P(B)
- 1
- P(A)

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Question # 2 of 10 (Start time: 08:50:03 AM, 04 September 2022)

A tree is normally constructed from______.

Select the correct option

- [ ] right
- [ ] right to left
- [ ] center
- [x] left to right

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MTH202 - Discrete Mathematics (Quiz No.4)

Question # 4 of 10 (Start time: 08:50:39 AM, 04 September 2022)

$P(A' \cap B') = ?$

Select the correct option

- ○ $P(A \cup B)$
- ● $P(A \cup B)'$
- ○ $P(AB)'$
- ○ $P(A \setminus B)'$

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Question # 5 of 10 (Start time: 08:51:35 AM, 04 September 2022)

If A and B are disjoint finite sets then

$$n(A \cup B) = \_\_\_\_\_\_.$$

Select the correct option

- [x] $n(A) + n(B)$
- [ ] $n(A) + n(B) - n(A \cap B)$
- [ ] $n(A) + n(B) + n(A \cap B)$
- [ ] $n(A) - n(B)$

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اس کے دیکھنے والوں کی سب مشکلیں آسان ہو جائیں گی صبح اس کو دیکھ کر جو کام شروع کیا جائیگا، پورا ہوگا، اور غیبی طریقوں سے رزق آنے لگے گا!